Why the head looks before it turns

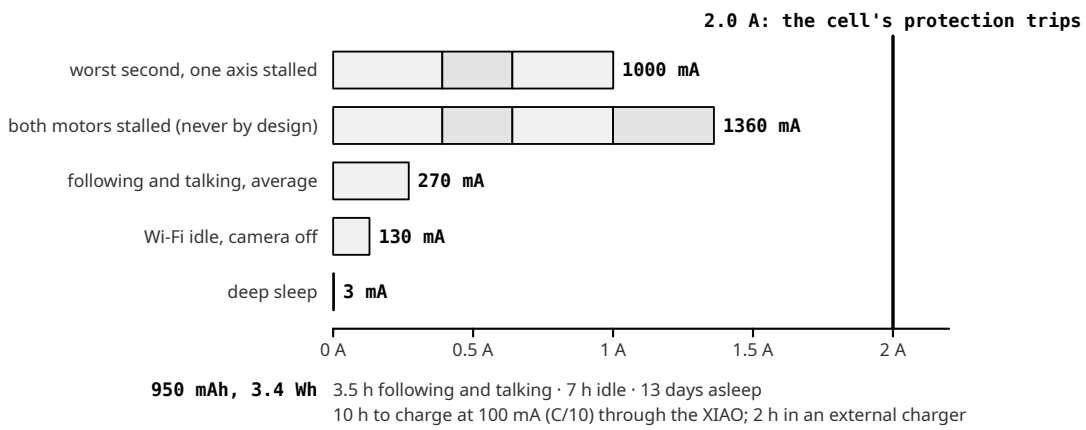
The working behind Rev B, approved 2026-11-10: one cell and no boost, a lens and a screen that share a head, two gearboxes on one bridge chip, a head balanced on its axis in a brass yoke, three buttons on one ADC pin, a clock line on the pin that chatters at boot, an amplifier that must be told to sleep, and a frame that floats. Read after Pocket AI with Camera, the sheet every number here comes from. Written 2026-11-10. Nothing measured yet. Source of truth: `pocket_ai_assistant_with_camera/docs/EDU.md`.

One cell, and why nothing is boosted

3.4 Wh is what the Nitecore cell holds: 950 mAh at 3.6 V nominal. A lithium-ion cell sits at 4.2 V full and is empty at 3.0 V, and every part on the sheet works across that whole range: the XIAO regulates it down to 3.3 V itself, the amp runs from 2.5 V up, the driver from 2 V up, the motors are 3 V parts.

A 5 V boost converter would have cost about 12 % of every watt as heat, plus its idle draw, plus a board and a rail to route. On a 3.5 hour run that is 25 minutes thrown away for nothing the parts asked for. So the cell is the only rail.

The cell is a protected one: a small board inside the can opens the circuit above 2.0 A, below 2.5 V and above 4.25 V. That is the last line, not the first: the design's worst second asks 1.00 A, and even two stalled motors at once would be 1.36 A.



The bars are the sheet's current budget; the dashed line is the protection's trip. Average running figures give 3.5 hours of following and talking, 7 idle and 13 days asleep.

Charging is slow on purpose. The XIAO's charger pushes 100 mA, which is C/10: 10 hours from empty, and only when the device sleeps, because awake it draws more than that. A charger follows a constant current until the cell reaches 4.20 V, then holds the voltage while the current tapers; the protected NL169 is built for exactly that profile. Below 0 °C lithium plates onto the anode as metal instead of entering it, which is why the rule says never charge a cold cell.

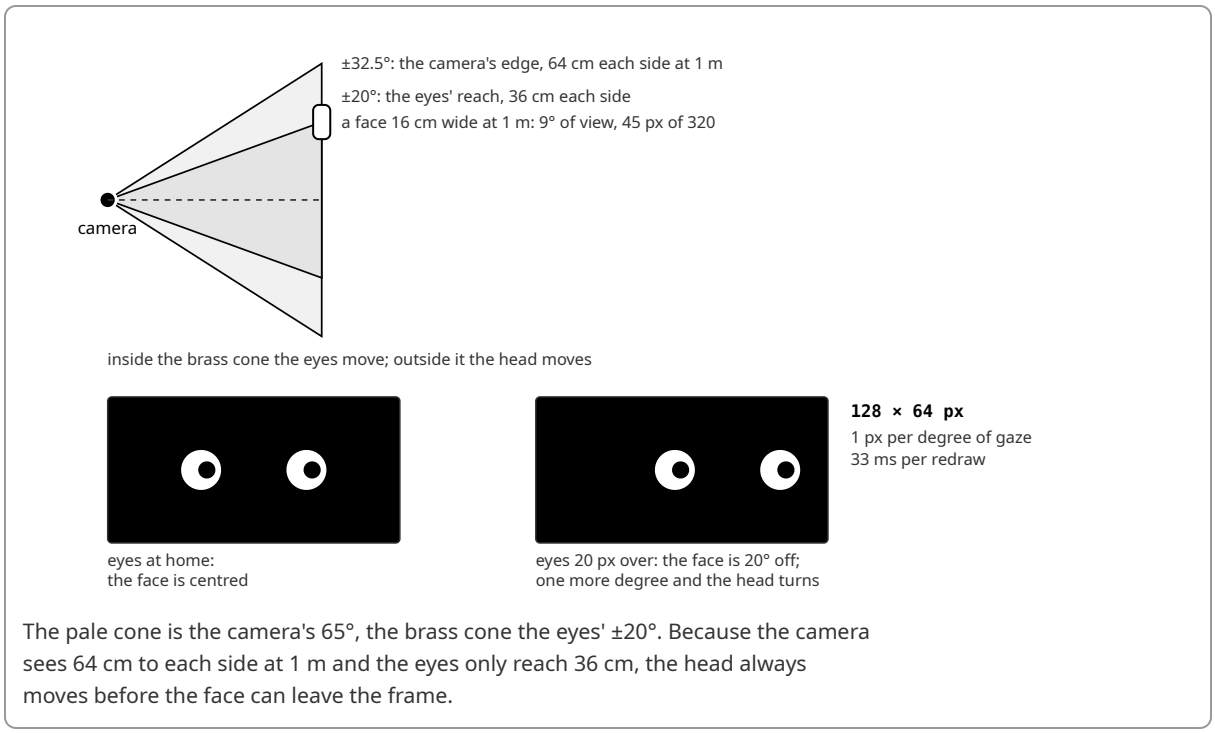
The camera, the eyes, and why the eyes move first

65° is what the camera's lens takes in across the frame. A lens is a pinhole with a focal length: the sensor is 3.52 mm wide, so the lens sits 2.76 mm behind it, and every pixel of a 320-wide frame covers 0.20°. A face 16 cm across

at 1 m spans 9°, about 45 pixels: plenty for a face finder, and enough to place its centre to a degree.

The eyes are drawn on 128 × 64 pixels of glass 21.7 mm wide. An eye can slide 20 pixels from home before it runs out of white, and the firmware lets that stand for 20° of gaze: 1 pixel per degree. The screen redraws every 33 ms, so the eyes follow a face the instant the frame says where it is.

The head is slower on purpose. At 16 rpm it turns 96° a second, so the 20° the eyes cover takes it 208 ms and a quarter turn 0.9 s. Fast eyes, slow head is how animals do it too: the eyes jump, then the head catches up and the eyes drift back to centre. The rule in the firmware is one line: while the face is inside ±20°, move the eyes; past it, move the head until it is centred.



Camera and screen share one head, so where the camera points is where the eyes point, by construction. The difference the builder asked about, the screen looking somewhere the camera is not, is the pixel offset of the eyes, not a second gimbal.

Two gear motors, and what a gearbox buys

A brushed DC motor spins at a speed set by its voltage and pushes with a torque set by its current. Bare, a 12 mm motor turns thousands of times a minute with almost no torque. The gearbox trades that speed for force: 16 rpm at the output means a few hundred turns of the motor for one of the shaft, and the torque is multiplied by the same ratio, less friction.

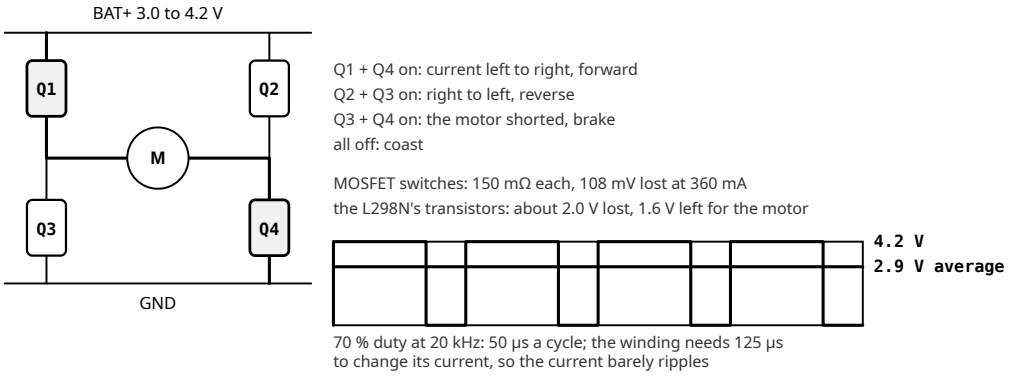
16 rpm is 96° a second, so a pulse of 100 ms is 9.6°. That is what makes timed moves repeatable without an encoder. It is also why the same gearbox holds the head where it was left: back-driving a high ratio means pushing the motor through all its stages of friction, which the head's weight cannot do when it sits on its axis.

The motors are 3 V parts on a cell that reads up to 4.2 V. The driver's PWM fixes that: on for 70 % of each cycle, off for the rest, and the winding's own inductance smooths the pulses into an average of 2.9 V at a full cell. At 20 kHz a cycle is 50 µs; the winding takes about 125 µs to change its current, so the ripple is small, and 20 kHz is above what an ear picks up.

Stall is the figure that matters for the wires and the cell: a blocked 12 mm motor draws about 360 mA. The home stops are what stall it on purpose for an instant at boot, and the firmware moves one axis at a time so the budget only ever carries one.

The H-bridge, and why the MX1508 and not the L298N

Reversing a motor means reversing which end of it sees the battery. An H-bridge is four switches around the motor; close one upper and the opposite lower and current flows one way, close the other pair and it flows the other. Close both lowers and the motor is shorted across itself, so its own back-EMF brakes it; open all four and it coasts.



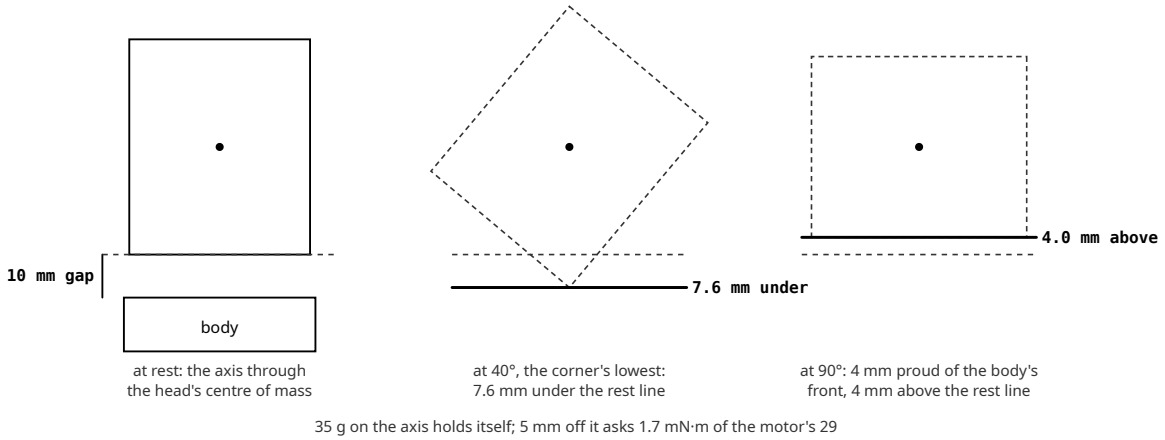
Forward lit. Below, the PWM: 70 % of 4.2 V averages 2.9 V.

The MX1508's switches are MOSFETs. On, a MOSFET is a resistor of about 150 mΩ, so two in the path lose 108 mV at 360 mA. The L298N the listing name borrows from uses bipolar Darlington transistors, which lose a fixed 2.0 V or more regardless of current: from a 3.6 V cell the motor would see 1.6 V. That is the whole reason one works on a cell and the other does not.

Two pins per motor, four for two: IN1 and IN2 for bridge A, IN3 and IN4 for bridge B, one chip. Each input has a 10 kΩ pull-down because a CMOS input left floating while the processor boots can read as high, and a motor that twitches at power-on is a motor that drifts off its home. GPIO44 wakes with a weak internal pull-up of about 45 kΩ; against the 10 kΩ that is 0.6 V at the driver, under its 1.8 V threshold.

Balance, the yoke, and the gap

The tilt axis runs through the head's centre of mass. A body balanced on its axis asks no torque to hold at any angle, so the tilt motor only ever works against friction and inertia. Miss the centre by 5 mm and 35 g asks 1.7 mN·m; the gearbox gives about 29 mN·m, enough to lift the whole head 86 mm off axis. Balance is a courtesy to the battery, not a requirement.



A 50 × 42 mm head turning about its centre. The lowest corner is at 40° of tilt, 7.6 mm under the rest line, which is why the gap to the body is 10 mm.

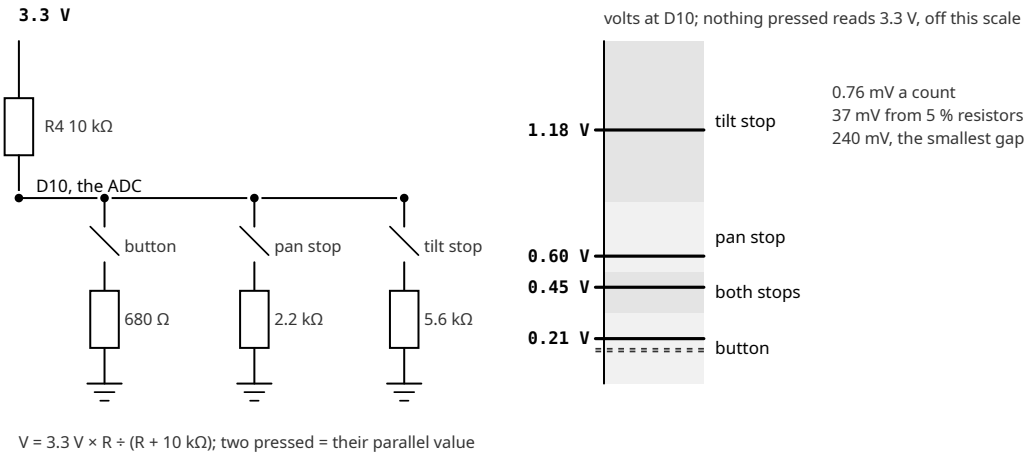
The yoke is a U of 1.5 mm brass tube because the head needs two bearings on one line and the pan hub under its middle. One arm carries the motor's D-shaft in a 4 mm tube socket, the other a plain 1.5 mm pin in a 3 mm tube; the motor side locates the head and the pin side only supports it, so nothing is over-constrained and nothing binds. The pan motor turns the yoke, head and all, from the hub below.

Wires that cross a joint are the part that wears. A bare rod bends a few hundred times before it cracks; fine silicone-insulated stranded wire bends hundreds of thousands. Five of them run up one upright as a single line, coiled once round the pan hub so ±150° of turn is spread along the coil, and looped at the tilt pin for the ±90°.

The ladder: three buttons on one pin

The XIAO has eleven header pins and the sheet needs twelve signals. The three switches became one by sharing an ADC pin through a resistor ladder. A 10 kΩ resistor pulls the pin to 3.3 V; each switch, when pressed, pulls it toward ground through its own resistor. The pin voltage is 3.3 V × R ÷ (R + 10 kΩ): 0.21 V for the button's 680 Ω, 0.60 V for the pan stop's 2.2 kΩ, 1.18 V for the tilt stop's 5.6 kΩ.

Two pressed at once put their resistors in parallel, which is always less than either, so the voltage drops further: 0.45 V for both stops, and anything with the button under 0.21 V. Every combination lands in its own place, and the firmware reads bands, not values: under 0.33 V is the button, 0.33 to 0.52 both stops, then the pan stop, then the tilt stop up to 1.6 V, and anything near 3.3 V is nothing.



The circuit and its scale. The smallest gap between neighbours is 240 mV; 5 % resistors move a reading by up to 37 mV and the ADC by about 30 mV. The generator asserts that margin.

The ADC has 4096 steps across about 3.1 V, 0.76 mV each, so the gaps are hundreds of steps wide. The 100 nF capacitor with the 10 kΩ source makes a 1 ms filter: the motors' 20 kHz switching is cut 126-fold before it reaches the pin, and a finger's bounce is averaged away. The ladder never draws more than 0.3 mA, and only while something is pressed.

The button is read in the same bands, so it is ignored for the second the head spends homing: a press then would look like a stop closing early. After that the ladder behaves as three separate buttons would.

Why the screen's clock sits on the UART pin

GPIO43 is the ESP32-S3's UART0 transmit pin, and the boot ROM prints its log there before any firmware runs: a burst of bits at 115200 baud, 8.7 µs each, for a fraction of a second. Rev A left the pin empty for that reason. Rev B needed it.

I2C has a grammar. A transaction begins only with a start condition: the data line falling while the clock line is high. Clock pulses on their own, with the data line sitting high on its pull-up, are not a start, so the screen's controller ignores them entirely. That makes SCL the one signal that boot chatter cannot disturb, and the S3's GPIO matrix lets I2C live on any pin, so it moved there. The console goes over USB, so the log is not missed.

The I2C bus itself runs at 400 kHz, 2.5 µs a bit, both lines open-drain with pull-ups, which is why neither line can ever fight the chatter: nothing on I2C drives high.

I2S in, I2S out, and why the amp must be told to sleep

The microphone on the Sense board speaks in single bits: a PDM stream at 2.048 MHz, 128 pulses for every sample, whose density is the sound pressure. The S3's I2S peripheral filters and decimates that into 16-bit samples at 16 kHz, speech bandwidth, with no analogue stage anywhere.

Out the other side the samples go to the amp as I2S: a bit clock at 512 kHz, a word clock at 16 kHz and the data line. The MAX98357A turns each sample into a pulse-width pattern at a few hundred kilohertz and switches the speaker between the rails with it; the speaker's own inductance and mass recover the audio and the switches waste almost nothing, which is why a class-D chip the size of a grain feeds a speaker from a cell.

The volume cap of 65 is a physics number. At a full cell the output swings 2.73 V peak, which into 8 Ω is 0.47 W, under the speaker's 0.8 W. Without the cap it would be 1.1 W, and the cone would not last the afternoon. The amp draws about 130 mA at that level, 341 mA at the peaks.

The shutdown line stays for a reason the datasheet's maker spells out: the amp must be shut down before its clocks stop. A class-D stage that loses its clock mid-pattern can leave the speaker held at DC, and 2.7 V of DC across 8 Ω is 0.93 W of pure heat in a voice coil rated for 0.8 W of music. So SD goes low first, then the clocks, every time, and the pull-down holds it low from the first microsecond of power.

What the Sense board owns, and the pin arithmetic

The camera talks over a parallel bus: eight data lines, a pixel clock, a frame sync, a line sync, the clock the camera is fed, and two lines of its own I2C-like control bus. Fourteen pins, soldered on the expansion board, not negotiable. The microphone takes two more. The micro-SD slot shares three of the header pins with its SPI, which is why the slot stays empty: those three drive the pan motor and the ladder.

That leaves the eleven header pins: three for I2S out, one for the amp's shutdown, two for I2C, four for the two bridges, one for the ladder. Eleven for eleven. Any future addition goes on the I2C bus, where an address costs no pins.

Brass, bare, and why the frame floats

Brass solders with the same iron and the same tin as the circuit, so the frame and its sockets are built on the bench with no other tool. It is left bare under the builder's rule: no paint, no skins. Brass does not rust; it darkens to a patina over months, which the family's other builds already wear.

Brass also conducts, and that is the one danger of a metal frame. A single touch between a rod and the frame is harmless; a second touch anywhere else makes a circuit the sheet never drew. So the frame connects to nothing, every rod keeps 2 mm from it, and the meter checks for continuity between the frame and both rails before the cell goes in. A floating frame cannot short, cannot carry motor current and cannot become an antenna for the Wi-Fi.

The parts that come in colours the rule does not allow are placed, not hidden: the red or blue driver board behind the cell holder, the amp under the speaker, the OLED's board behind its own glass. Light from the device is the screen's white pixels and nothing else.

The numbers, and where they come from

NUMBER	WHAT IT IS
`2.76 mm`	the lens's focal length: a 3.52 mm sensor seeing 65°
`0.20° per pixel`	65° over a 320-pixel frame
`45 px`	a 16 cm face at 1 m in that frame
`1 px per degree`	20 px of eye travel standing for 20° of gaze
`96° per second`	16 rpm at the gearbox output
`9.6°`	one 100 ms pulse of the motor
`2.9 V`	70 % duty at a 4.2 V cell
`108 mV`	lost in the MX1508's two switches at 360 mA stall
`1.6 V`	what an L298N would leave the motor from a 3.6 V cell
`1.7 mN·m`	holding a 35 g head 5 mm off its axis; the motor has about 29

`7.6 mm`	how far the head's corner dips below its rest line, at 40°
`0.21 / 0.60 / 1.18 / 0.45 V`	the ladder: button, pan stop, tilt stop, both stops
`0.76 mV`	one ADC count: 3.1 V over 4096
`1 ms`	the ladder's RC filter: 10 kΩ × 100 nF
`512 kHz`	the I2S bit clock: 16 kHz × 16 bits × 2 channels
`2.048 MHz`	the microphone's clock: 128 × 16 kHz
`0.47 W`	into 8 Ω at the volume cap of 65 and a full cell; the speaker is rated 0.8
`0.93 W`	the heat if the amp were ever left at DC: why SD goes low before the clocks
`3.4 Wh`	the cell: 950 mAh × 3.6 V
`1.00 A`	the worst second, against a 2.0 A trip
`3.5 h · 7 h · 10 h`	following and talking · Wi-Fi idle · charging through the XIAO

Which part each idea lives in

- **the lens, the sensor and the microphone:** Seeed XIAO ESP32-S3 Sense
- **the H-bridges:** MX1508 dual motor driver, Aideepen
- **the gearboxes:** 12 mm gear motor, 3 V 16 rpm, Naroote
- **the cell:** Nitecore NL169 16340, protected
- **its holder:** 16340 / CR123A holder with leads, ACEIRMC
- **the switch in the cell's lead:** MTS-102 toggle, Ferwooh
- **the eyes:** 0.96 inch SSD1306 OLED, Hosyond
- **the class-D stage:** MAX98357A I2S amplifier, HiLetgo
- **the voice coil:** Same Sky CES-20134-088PM enclosed speaker
- **the ladder's three switches:** tactile buttons, QTEATAK
- **the ladder's resistors and the pull-downs:** resistor kit, LuminologyPro
- **the yoke and the frame:** K&S 1.5 mm brass tube and rod
- **the shaft sockets:** METERXITY brass tube set, 2 to 7 mm

What comes next

The controller pair is the only part not yet in the building; Amazon's estimate is this week. Everything on this page becomes a measurement at the bench in the order the sheet's checks give: the ladder's five voltages with a meter before power (0.21, 0.60, 1.18, 0.45 V and 3.3 V), each motor's running and stall current with the head off the shafts (under 0.5 A stalled is the pass), the camera stream's draw (under 400 mA), the first charge watched with a meter in the cell's lead (about 100 mA), and the speaker scoped differentially under 2.7 V peak. Firmware is the step after that, and the first of it is the XIAO alone on USB with the screen, exactly as the first Pocket AI began. The first measurements replace this page's estimates: the head's mass and balance, the stall current, the ladder's real voltages, the hours per charge.

Sources

- Seeed wiki: Getting started with XIAO ESP32S3 (Sense): pinout, camera and microphone GPIOs, power figures
- Seeed wiki: XIAO ESP32S3 Sense camera usage (OV2640, the expansion board)
- OV2640 datasheet, OmniVision: 1/4 inch sensor, 2.2 µm pixels
- ESP32-S3 Technical Reference Manual: ADC (chapter 39), GPIO matrix (chapter 6), I2S (chapter 32)
- ESP-IDF, ADC oneshot driver for the ESP32-S3: attenuation and range
- MAX98357A/B datasheet, Analog Devices (Maxim): SD_MODE, output power, shutdown
- ADI EngineerZone: shut the MAX98357 down before removing LRCLK
- L298N datasheet, STMicroelectronics: saturation voltages
- MX1508 listing: 2-10 V, 1.5 A, 1.8-7 V inputs
- I2C-bus specification UM10204, NXP: the START condition
- CES-20134-088PM datasheet, Same Sky: 8 Ω, 0.8 W
- Nitecore NL169 at IMR Batteries: 950 mAh, protected, 2 A
- Pocket AI with Camera, Rev B build page: the sheet these numbers come from

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Built by `docs/pages/gen/edu_page.py`, which also writes `docs/EDU.md`. Assumptions, each a number at the top of the script: a 65° lens; a 16 cm face at 1 m; 20 px of eye travel for 20°; a 35 g head; 360 mA stall and 0.3 kg·cm for the 12 mm motor class; 150 mΩ switches in the MX1508; a 45 kΩ internal pull-up; 85 % amplifier efficiency; 100 mA of charge; a boost converter at 88 % had there been one. The rest is datasheet or the sheet. Nothing here has been measured on the build.